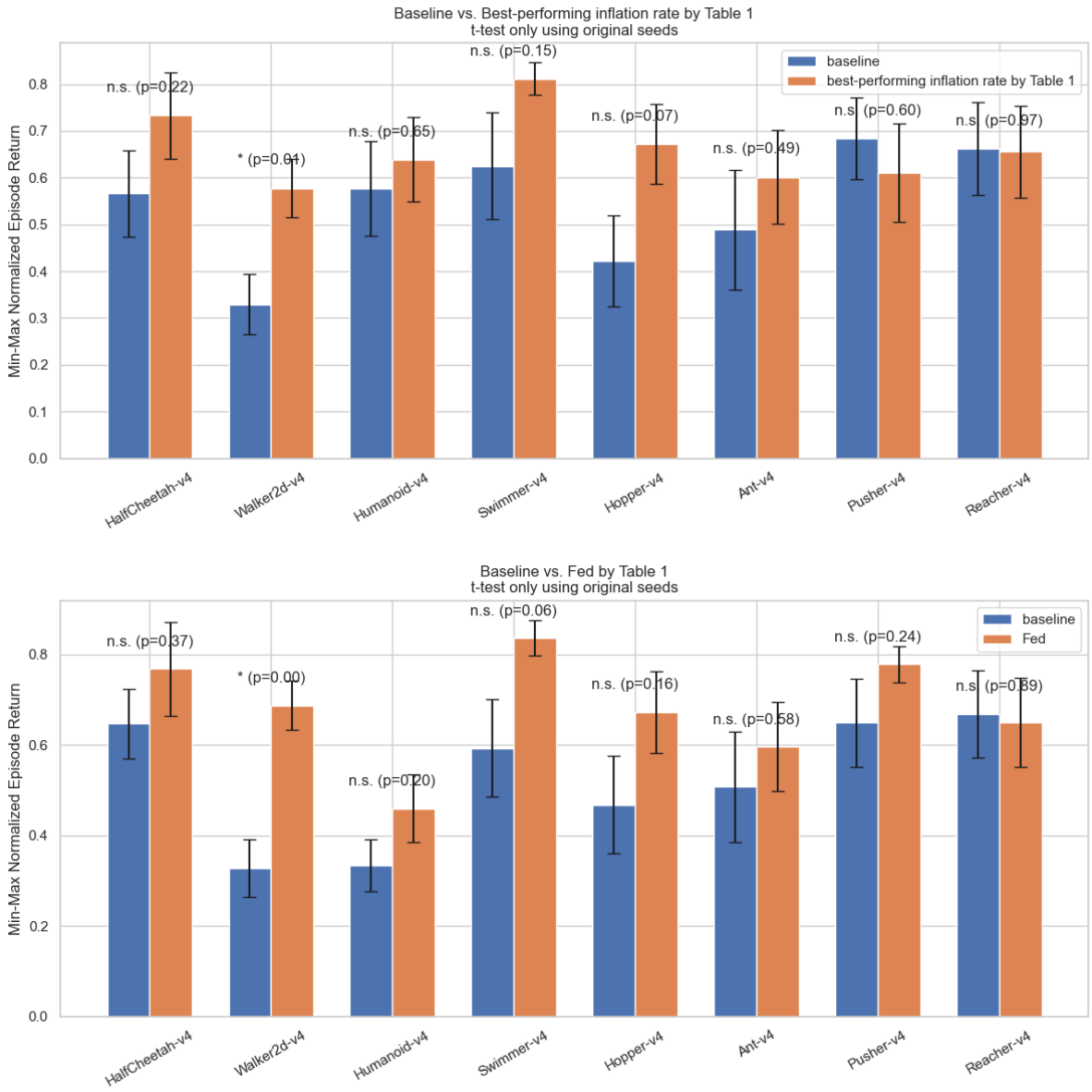

Figure 1: Per-task t-tests across MuJoCo tasks using the original 10 seeds. (Top) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Bottom) Baseline vs. Fed.

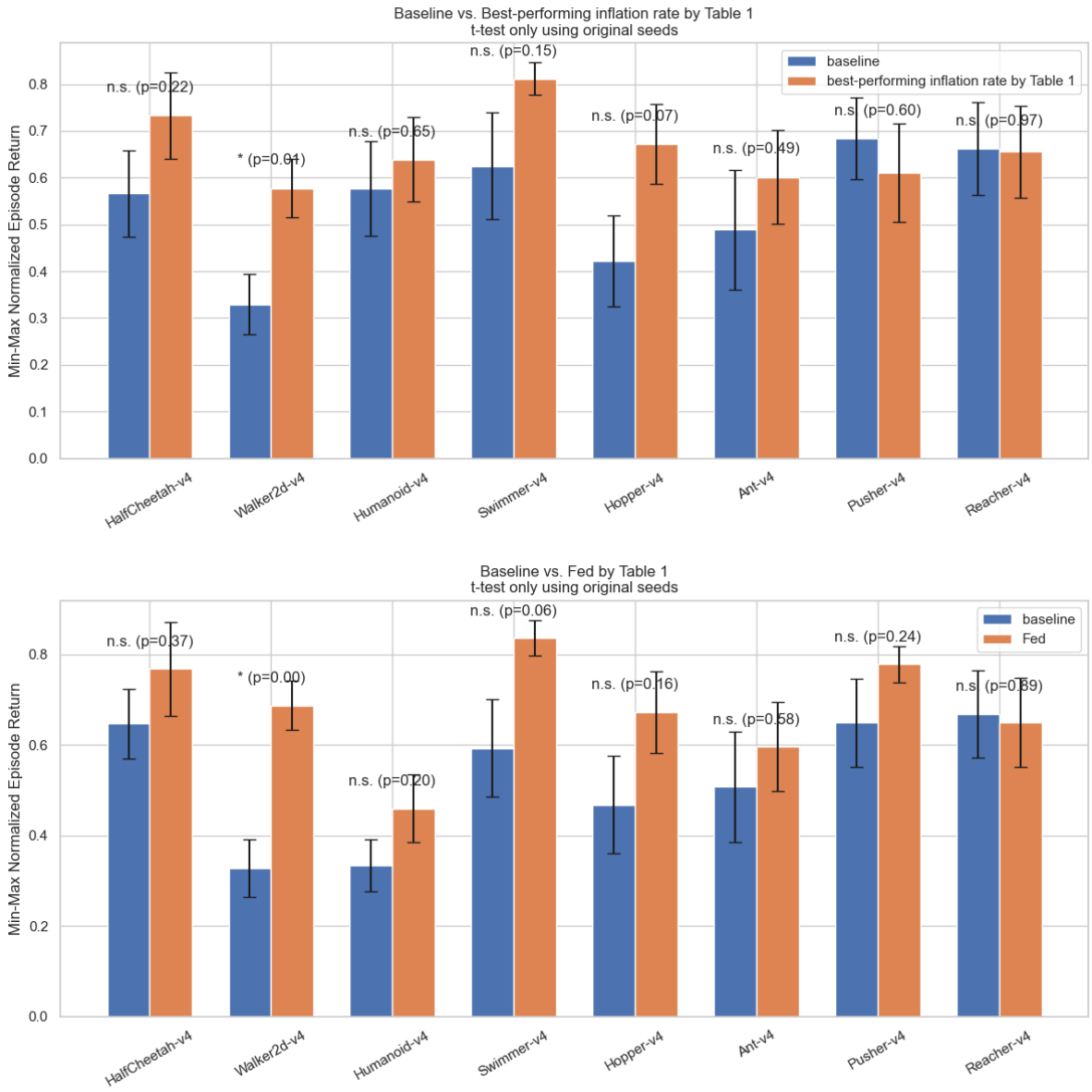

Figure 2: Per-task t-tests across MuJoCo tasks using additional seeds to gain more statistical power. (Top) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Bottom) Baseline vs. Fed.

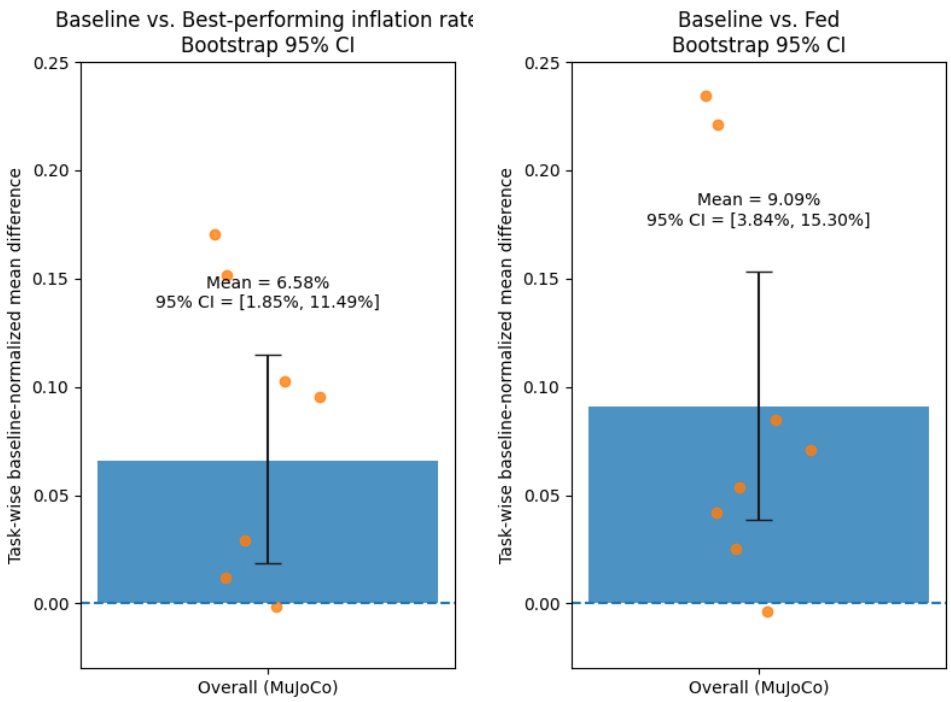

Figure 3: Task-wise bootstrap 95% CI over MuJoCo tasks. (Left) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Right) Baseline vs. Fed.

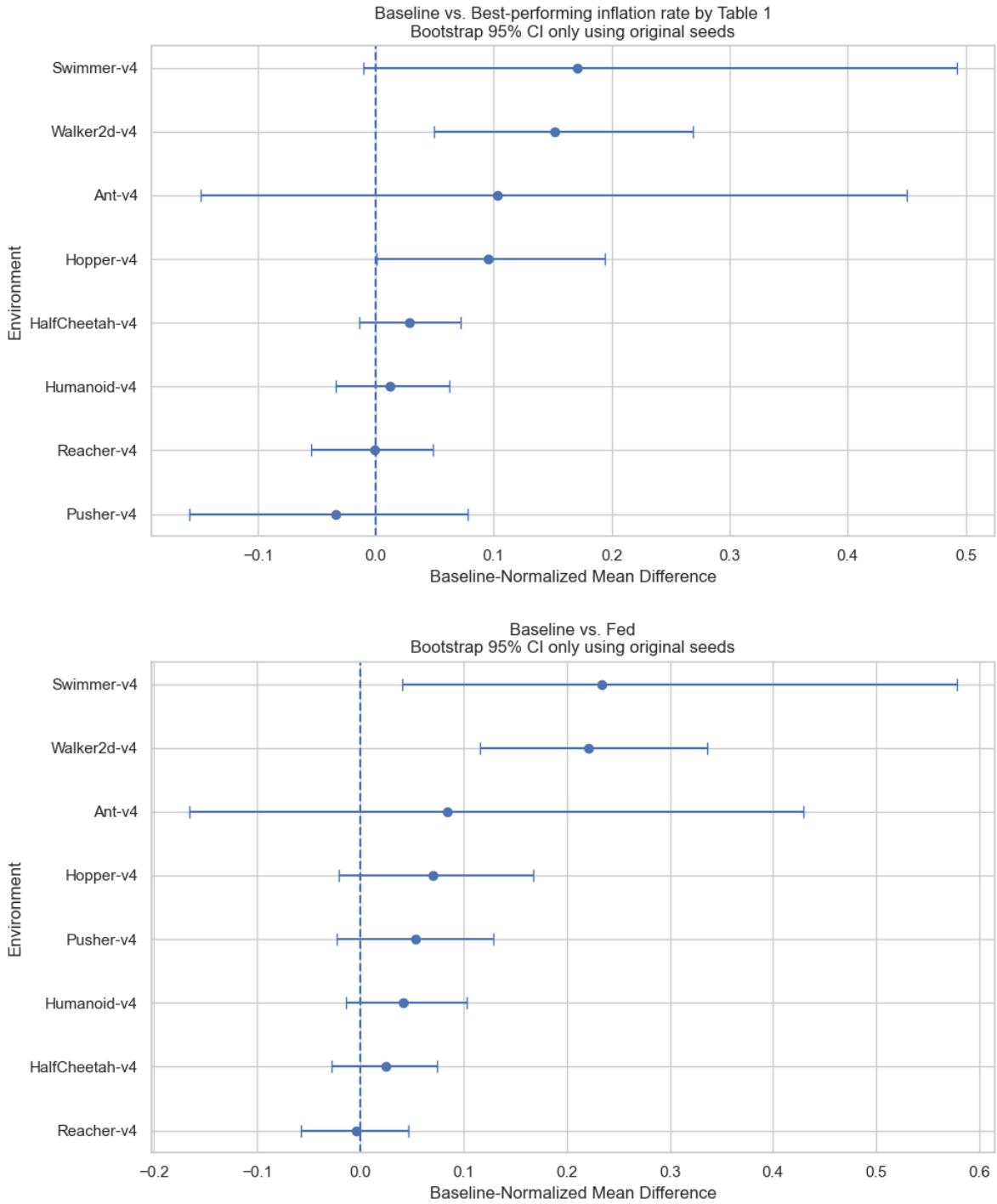

Figure 4: Per-task bootstrap 95% CI across MuJoCo tasks using the original 10 seeds. (Top) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Bottom) Baseline vs. Fed.

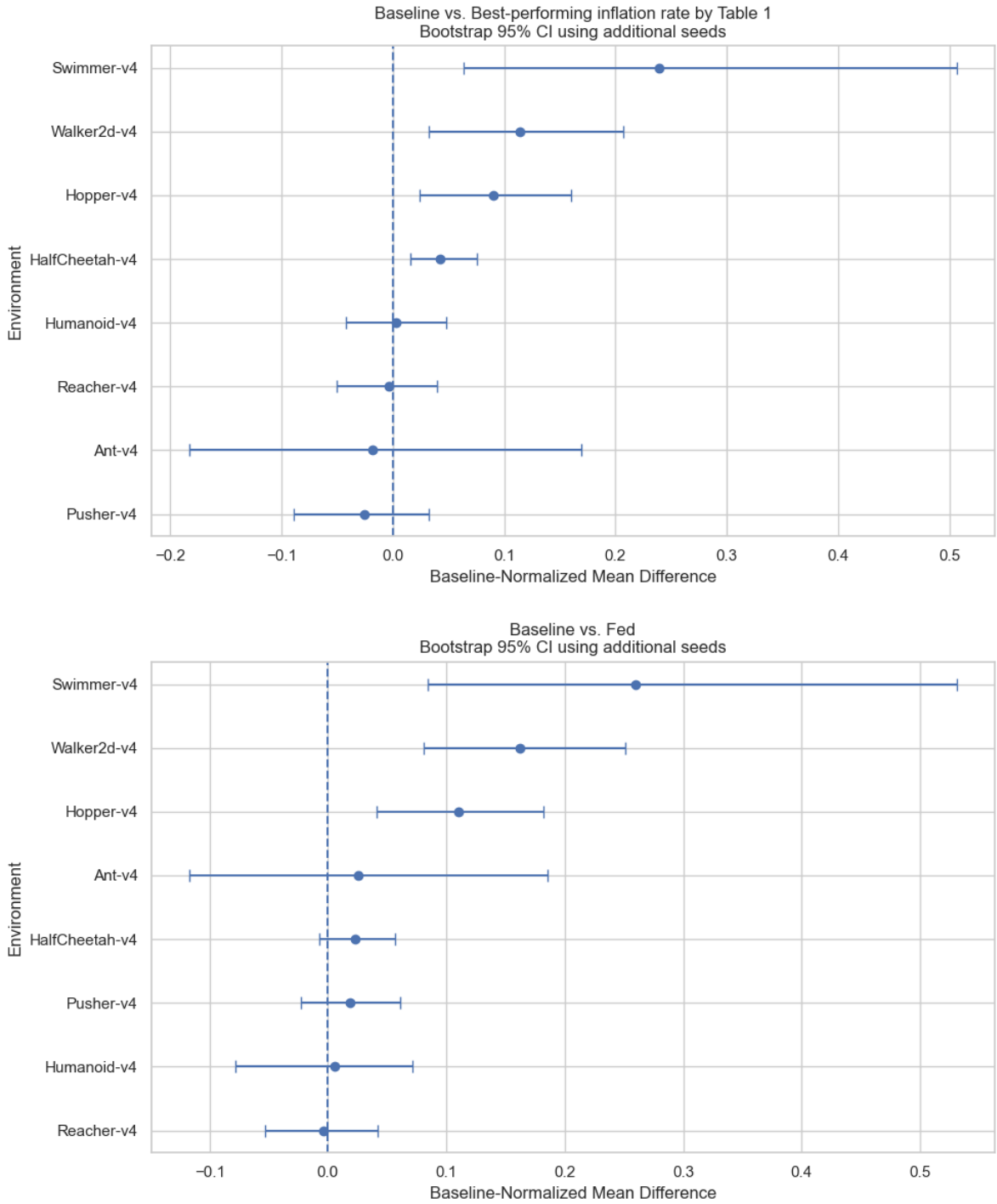

Figure 5: Per-task bootstrap 95% CI across MuJoCo tasks using additional seeds to gain more statistical power. (Top) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Bottom) Baseline vs. Fed.

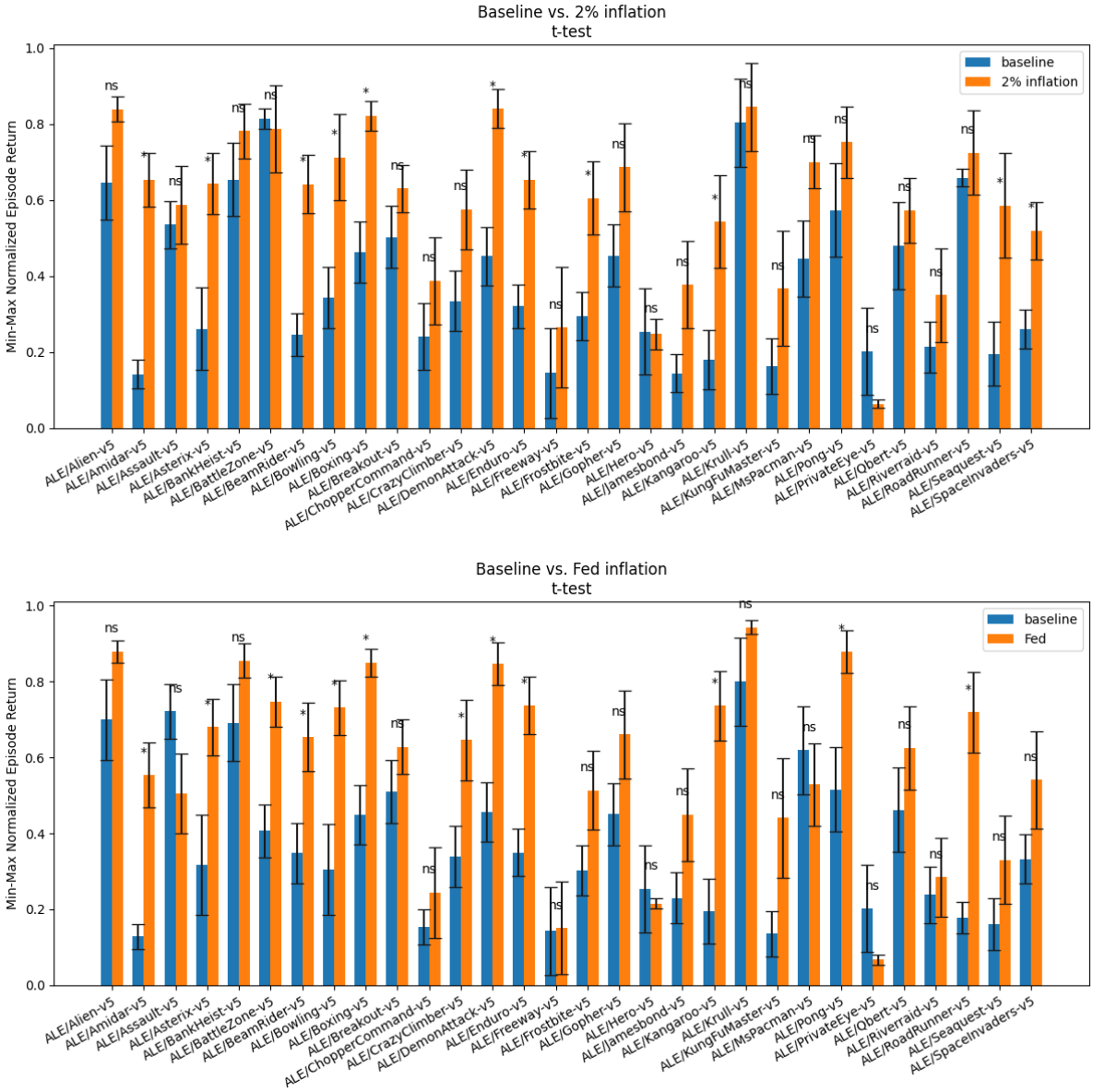

Figure 6: Per-task t-tests across Atari tasks. (Top) Baseline vs. 2% inflation. (Bottom) Baseline vs. Fed.

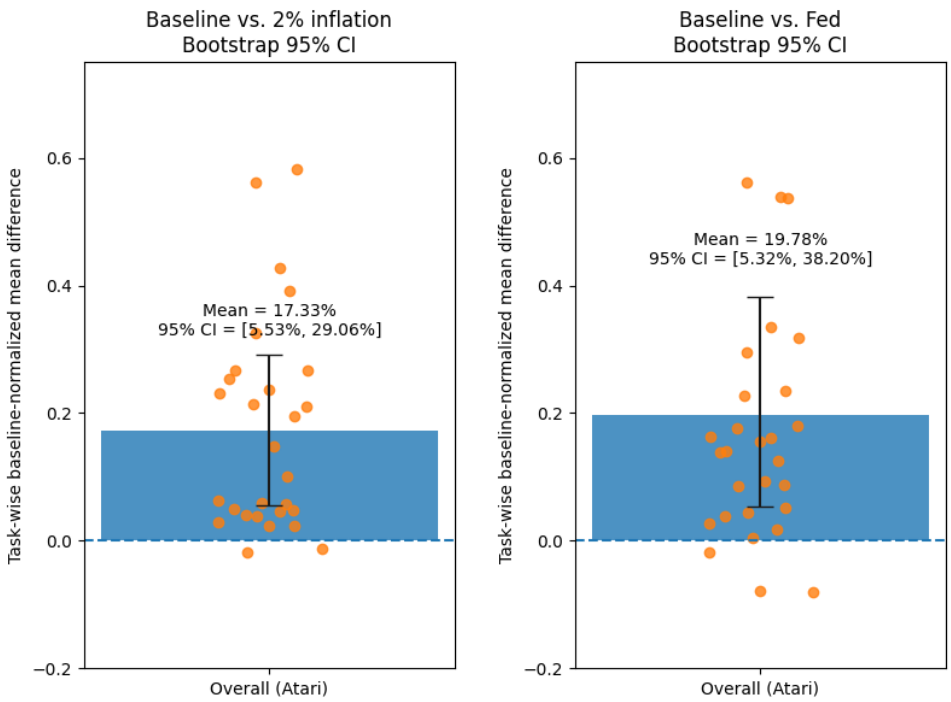

Figure 7: Task-wise bootstrap 95% CI over Atari tasks. (Left) Baseline vs. 2% inflation rate. (Right) Baseline vs. Fed.

Table 1: Performance comparison under different learning rate inflation settings. We added fixed inflation rates (5%, 10%) and the Fed-based adaptive learning rate inflation (lr-Fed) with a 2% target rate on.

LR Inflation				
Rate	**HalfCheetah**	**Walker2d**	**Humanoid**	**Swimmer**
0%	**15241±734**	4399±527	**5826±313**	116±36
1%	14860±1560	4729±422	5634±151	107±35
2%	14566±1006	**4922±515**	5417±298	77±41
3%	14643±895	4646±660	5282±226	90±41
4%	13854±905	4502±507	5302±280	113±30
5%	12902±1793	4508±472	4952±286	88±36
10%	9932±633	3548±597	2926±476	74±25
lr-Fed (2%)	14783±869	4875±682	5473±408	**102±40**

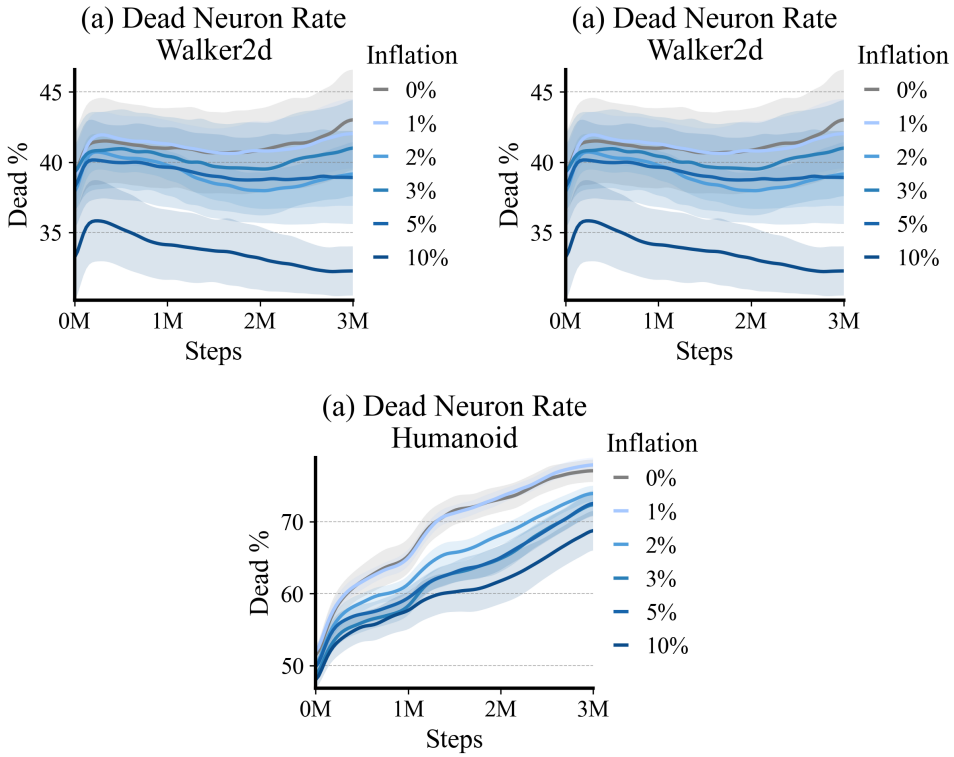

Figure 8: Additional results of dead neuron ratios across different tasks under varying reward inflation rates (Fig. 2(a) in the main paper corresponds to HalfCheetah-v4).